

Michael Zandonella

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EDUCATION

Oregon State University

Bachelor of Science in Computer Science | 3.96 GPA

Corvallis, OR

Expected June 2026

EXPERIENCE

Computer Science Peer Tutor

October 2025 – Present

College of Engineering, Oregon State University

Corvallis, OR

- Debugged and optimized implementations under time-constrained conditions, rapidly adapting to unfamiliar codebases and resolving ambiguous issues
- Solved complex data structures and algorithm problems by breaking down logic, identifying edge cases, and guiding solutions from approach to implementation
- Improved student understanding in Data Structures, Algorithms, and Web Development by delivering targeted 1-on-1 instruction to 5–15 students per term

Web Assistant

June 2024 - July 2025

OSU Beaver Store

Corvallis, OR

- Reduced manual back stock processing time by building an internal Python tool that scraped product images and catalog data to generate printable labels
- Maintained data quality and consistency across a large-scale e-commerce catalog by auditing thousands of listings and standardizing 500+ product assets
- Improved site consistency and usability by analyzing external reports and implementing changes to content structure and presentation

PROJECTS

Rotations.lol | TypeScript, React, Node.js, Supabase, PostgreSQL, AWS SES, Docker

- Built and operated a distributed data ingestion pipeline that collects and normalizes live sale and catalog data into PostgreSQL
- Enabled automated sale tracking and notifications for 100+ users by building a full-stack system processing 16,000+ items and ~80 weekly rotations
- Delivered reliable, deduplicated user notifications by designing a stateful system that matches wishlists against rotation data and handles email delivery via AWS SES and SNS

RFID-Based Asset Tracking System (Senior Capstone) | Node.js, React, Socket.IO, Express, SQLite, Docker

- Led system integration across frontend, backend, and scanner components by defining Socket.IO event contracts and collaborating with stakeholders to refine requirements and system design
- Architected and built a real-time asset tracking system with role-based access by developing an event-driven RFID scanner service using Node.js and WebSockets
- Unblocked parallel development for 4 teammates by building a mock scanner API and control interface to simulate hardware interactions without physical devices

YouTube Transcript Search | Python, Elasticsearch, React, Flask

- Improved search relevance and recall by implementing BM25-based retrieval with multi-field queries, phrase boosting, and fuzzy matching in Elasticsearch
- Enabled timestamp-level search across long-form video content by building a transcript ingestion pipeline that processes and indexes captions
- Reduced time to locate relevant content by building a React interface with direct timestamped video linking

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend & Frameworks: Node.js, React, Express, Flask

Databases & Search: PostgreSQL, MongoDB, Elasticsearch

Infrastructure & Tools: Docker, Kubernetes, Terraform, Ansible, AWS, Nginx, Linux, Git